

Operating Systems
Lab 10

1. From the BoundedBuffer class attached to this exercise. Complete the class by initializing the constructor to initialize all variables. Complete the “insert” method that a producer thread will call to insert data into the bounded buffer. This method also displays the producer thread's work, as shown by the example output. Complete the “remove” method that a consumer thread will call to read data from the bounded buffer. This method also displays the consumer thread's work, as shown by the example output.

Write a Producer class as a Thread class that inserts 20 items into the buffer (one at a time). Write a Consumer class as a Thread class that reads 20 items from the buffer (one at a time). Write a main program to create a BoundedBuffer object, a Producer object, and a Consumer object by sharing the Bounded buffer object via the constructors of the Producer and Consumer classes. Start the Producer and Consumer threads. Wait for both threads to complete their tasks. The example output of the program is as follows:

```
Producer Entered 0 Buffer Size = 1
Consumer Consumed 0 Buffer EMPTY
Producer Entered 1 Buffer Size = 1
Consumer Consumed 1 Buffer EMPTY
Producer Entered 2 Buffer Size = 1
Consumer Consumed 2 Buffer EMPTY
Producer Entered 3 Buffer Size = 1
Consumer Consumed 3 Buffer EMPTY
Producer Entered 4 Buffer Size = 1
Consumer Consumed 4 Buffer EMPTY
Producer Entered 5 Buffer Size = 1
Consumer Consumed 5 Buffer EMPTY
Producer Entered 6 Buffer Size = 1
Consumer Consumed 6 Buffer EMPTY
Producer Entered 7 Buffer Size = 1
Producer Entered 8 Buffer Size = 2
Consumer Consumed 7 Buffer Size = 1
Consumer Consumed 8 Buffer EMPTY
Producer Entered 9 Buffer Size = 1
Producer Entered 10 Buffer Size = 2
Producer Entered 11 Buffer Size = 3
Consumer Consumed 9 Buffer Size = 2
Producer Entered 12 Buffer Size = 3
Consumer Consumed 10 Buffer Size = 2
Producer Entered 13 Buffer Size = 3
Consumer Consumed 11 Buffer Size = 2
Consumer Consumed 12 Buffer Size = 1
Producer Entered 14 Buffer Size = 2
Producer Entered 15 Buffer Size = 3
Consumer Consumed 13 Buffer Size = 2
Producer Entered 16 Buffer Size = 3
Consumer Consumed 14 Buffer Size = 2
Consumer Consumed 15 Buffer Size = 1
Producer Entered 17 Buffer Size = 2
Consumer Consumed 16 Buffer Size = 1
Producer Entered 18 Buffer Size = 2
Consumer Consumed 17 Buffer Size = 1
Producer Entered 19 Buffer Size = 2
Consumer Consumed 18 Buffer Size = 1
Consumer Consumed 19 Buffer EMPTY
```